

Practice Paper B - Exam Solutions

1. (a) (i) Consider the vectors $\mathbf{u} = (1, 0, 2)$, $\mathbf{v} = (1, 3, 4)$ and $\mathbf{w} = (-3, 0, -2)$. Calculate the dot product $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w})$. (3 marks)

Solution: First calculate $\mathbf{v} + \mathbf{w}$:

$$\mathbf{v} + \mathbf{w} = (1, 3, 4) + (-3, 0, -2) = (-2, 3, 2)$$

[1 mark]

Then

$$\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = (1 \times -2) + (0 \times 3) + (2 \times 2) = -2 + 4 = 2$$

[2 marks]

- (ii) What would be true of two vectors whose dot product:

- is large and positive?
- is large and negative?
- is zero?

(3 marks)

Solution:

- They point in roughly the same direction
- They point in very different directions
- They are at right angles to each other/they are perpendicular.

[3 marks]

- (b) Consider the following system of linear equations:

$$\begin{aligned}x + 2y &= 1, \\4x + 9y + z &= -2, \\3x + 6y + z &= 2.\end{aligned}$$

- (i) Present the system of equations in the matrix form $A\mathbf{x} = \mathbf{b}$. Clearly denote the values of A , $\mathbf{b}$ and $\mathbf{x}$. (5 marks)

Solution:

$$\underbrace{\begin{pmatrix} 1 & 2 & 0 \\ 4 & 9 & 1 \\ 3 & 6 & 1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}}_{\mathbf{b}}$$

[5 marks]

- (ii) Using Gauss-Jordan elimination, find the inverse of the matrix A from part (i). (10 marks)

Solution:

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 4 & 9 & 1 & 0 & 1 & 0 \\ 3 & 6 & 1 & 0 & 0 & 1 \end{array} \right) \begin{array}{l} \\ R_2 - 4R_1 \\ R_3 - 3R_1 \end{array} \\ \rightarrow & \left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -4 & 1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array} \right) \begin{array}{l} \\ R_1 - 2R_2 \\ \\ \end{array} \\ \rightarrow & \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 9 & -2 & 0 \\ 0 & 1 & 1 & -4 & 1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array} \right) \begin{array}{l} R_1 + 2R_3 \\ R_2 - R_3 \\ \\ \end{array} \\ \rightarrow & \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & -2 & 2 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array} \right) \end{aligned}$$

Therefore $A^{-1} = \begin{pmatrix} 3 & -2 & 2 \\ -1 & 1 & -1 \\ -3 & 0 & 1 \end{pmatrix}$.

[10 marks; lose one for each error]

- (iii) Using this inverse matrix, and the vector b from part (i), find the solution to the system of equations. (5 marks)

Solution: Since $Ax = b$, we have $x = bA^{-1}$, so the solution will be given by:

$$\begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} \begin{pmatrix} 3 & -2 & 2 \\ -1 & 1 & -1 \\ -3 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -4 \\ 6 \end{pmatrix}$$

Hence the solution to the system is $x = -1$, $y = -4$ and $z = 6$.

[5 marks]

Total marks for question 1: [26]

2. (a) A proof-reader reads through a 250-page proof of a book. The numbers of mistakes found on each page are summarized in the table.

Number of mistakes	0	1	2	3	4
Number of pages	61	109	53	23	4

- (i) Determine the mean number of mistakes found per page. (4 marks)

Solution: The mean is the total number of mistakes divided by the number of pages, given by:

$$\frac{(61 \times 0) + (109 \times 1) + (53 \times 2) + (23 \times 3) + (4 \times 4)}{61 + 109 + 53 + 23 + 4} = \frac{300}{250} = 1.2$$

[4 marks]

- (ii) Find the variance in the number of mistakes found per page. (5 marks)

Solution: Since this is the entire population, the variance is given by:

$$\sigma^2 = \frac{1}{n} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

[1 mark]

There are 250 measurements, so $n = 250$. The sum of the values we calculated earlier to be 300.

[1 mark]

The sum of the squares is given by:

$$\sum x^2 = (61 \times 0^2) + (109 \times 1^2) + (53 \times 2^2) + (23 \times 3^2) + (4 \times 4^2) = 592$$

So using the formula:

$$\sigma^2 = \frac{1}{250} \left(592 - \frac{300^2}{250} \right) = 0.928$$

[3 marks]

- (iii) If this book is opened at a random page, what is the probability that page will contain no mistakes? (2 marks)

Solution: Of the 250 pages in the book, 61 contain no mistakes, so the probability is $\frac{61}{250} = 0.244$.

[2 marks]

- (b) The publisher wants a report on how many mistakes are made in their books generally, and asks the proof-reader to look at 100 different book proofs. The proof-reader suggests it will take too long to check them all, but that they could check a sample of pages instead.

- (i) What is a proportional stratified sample? (2 marks)

Solution: Consider the population split into strata, and take a sample from each proportional to the strata's proportion of the population.

[2 marks]

- (ii) Describe how you might take a proportional stratified sample of the pages in the books. Include details of how you might use software to do this. (6 marks)

Solution: Make a list, using a spreadsheet, of all the books the publisher would like to check, and how many pages each book contains. [2 marks]

For each book, calculate how many pages would be in a proportional sample, by taking a fixed proportion (say, 5%) of the total number of pages. [2 marks]

Use a random number function to generate a list, for each book, of that many random numbers between 1 and the number of pages in the book. These pages will constitute the stratified sample. [2 marks]

Total marks for question 2: [19]

3. (a) A scientist wishes to do a study comparing the strength people have in their left and right arms. The participants are recruited online, and asked to visit their local gym and complete as many bicep curls as they can using a heavy weight, first with one arm and then the other. They submit the number of curls they completed with each arm online.

(i) Comment on two potential sources of inaccuracy in this data. (4 marks)

Solution: Sources of accuracy might include:

- The weight to be lifted is not specified, so people may be lifting different weights
- Some people may be willing to endure more pain than others so may complete more reps before tiring
- If people use one particular arm first, it may tire them out so they can do fewer reps with the second arm
- The study is self-reporting, so people may lie on their response
- Any other reasonable suggestion

[4 marks (2 per suggestion, 1 if underexplained)]

A sample of the data reported by the participants is given below.

Left	14	15	26	15	20	23	21	7	11	25	7	9	25
Right	23	16	28	19	19	24	32	12	13	33	14	15	28

- (ii) Assuming the differences between the left and right arms follow a normal distribution, perform a paired t-test at the 5% significance level to compare the left and right arm data from the participants. (15 marks)

Solution: Let L be the number of reps completed using the left arm, and R be the number of reps completed using the right arm.

Let $D = R - L$, which we assume is normally distributed. [1 marks]

$H_0: \mu_D = 0$; $H_1: \mu_D \neq 0$ [2 marks]

Use a two-tailed paired t -test at the 5% level using $\nu = 13 - 1 = 12$. [2 marks]

The differences between the pairs of values are 9, 1, 2, 4, -1, 1, 11, 5, 2, 8, 7, 6 and 3. We can calculate the mean:

$$\bar{d} = \frac{9 + 1 + 2 + 4 + (-1) + 1 + 11 + 5 + 2 + 8 + 7 + 6 + 3}{13} = \frac{58}{13} = 4.46$$

[2 marks]

and standard deviation:

$$s_d = \sqrt{\frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)}$$

We know $\sum x = 58$ from earlier, and $n = 13$. To find the sum of squares:

$$9^2 + 1^2 + 2^2 + 4^2 + (-1)^2 + 1^2 + 11^2 + 5^2 + 2^2 + 8^2 + 7^2 + 6^2 + 3^2 = 412$$

Then

$$s_d = \sqrt{\frac{1}{12} \left(412 - \frac{58^2}{13} \right)} = \sqrt{12.76} = 3.57$$

[3 marks]

So the test statistic is:

$$t = \frac{\bar{d} - \mu}{\frac{s_d}{\sqrt{n}}} = \frac{4.46 - 0}{\frac{3.57}{\sqrt{13}}} = 4.5$$

[2 marks]

The critical value for a two-tailed test with $\nu = 12$ is 2.179, so the critical region will be values $t < -2.179$ and $t > 2.179$. [2 marks]

The test statistic is in the critical region, so we can reject the null hypothesis and conclude that there is a difference in the mean strength between left and right arms. [1 mark]

(b) The regression line for this data is given by

$$y = 5.91 + 0.913x$$

The PMCC is 0.876.

- (i) Based on the equation for the regression line and the value of the PMCC, describe the type of correlation observed. (3 marks)

Solution: The line has a positive gradient, so there is a positive correlation between the two variables. The PMCC is greater than 0.7, so we could categorise this as a very strong positive correlation. [3 marks]

- (ii) Calculate the residual for the first participant in the study. (3 marks)

Solution: The first participant has $L = 21$, $R = 32$. For this value of L , the value on the regression line is given by:

$$y = 5.91 + 0.913 \times 21 = 25.083$$

The residual is then

$$32 - 25.093 = 6.917$$

[3 marks]

- (iii) Use the regression line to predict how many reps someone could do with their right arm if they could do 18 with their left arm. (2 marks)

Solution: If someone had $L = 18$, the value on the regression line would be given by:

$$y = 5.91 + 0.913 \times 18 = 22.344$$

[2 marks]

Total marks for question 3: [27]

4. (a) An ice cream vendor sells two flavours of ice cream, and for each flavour, the number of sales each day over the season is normally distributed. Let C represent the sales of chocolate ice cream, and V the sales of vanilla ice cream. The vendor has observed that

$$C \sim N(85, 11^2) \quad \text{and} \quad V \sim N(93, 7^2)$$

- (i) Write down a model for the total number of ice creams sold in a day, $C + V$, and explain why it is valid. (3 marks)

Solution: The sum of two normally distributed variables is also normal. [1 mark]

We have $C + V \sim N(85 + 93, 11^2 + 7^2) = N(178, 13.04^2)$ [2 marks]

- (ii) What is probability that more than 200 ice creams are sold in total on a given day? (5 marks)

Solution: For the value 200, the z -score is given by $z = \frac{x - \mu}{\sigma} = \frac{200 - 178}{13.04} = 1.69$ [2 marks]

Looking this up in the table gives a probability of 0.9545. This is the area to the left of 200, so to find the probability more than 200 ice creams are sold we need to calculate

$$1 - 0.9545 = 0.0455 = 4.55\%$$

[3 marks]

- (iii) The price of a chocolate ice cream is £2.80, and a vanilla ice cream is £1.90. Write down a model X for the total money taken in pounds, and explain why it's a valid model. Explain what the model tells you about how much money the vendor might expect to take in a day, using the standard deviation to make a prediction about how that might vary. (9 marks)

Solution: The amount of money taken can be written as

$$X = 2.8C + 1.9V$$

[1 mark]

A linear combination of normal distributions is also a normal distribution, and we have:

$$2.8C \sim N(2.8 \times 85, 2.8^2 \times 11^2) = N(238, 30.8^2)$$

$$1.9V \sim N(1.9 \times 93, 1.9^2 \times 7^2) = N(176.7, 20.9^2)$$

[3 marks]

$$\Rightarrow 2.8C + 1.9V \sim N(238 + 176.7, 30.8^2 + 20.9^2) = N(414.7, 37.22^2)$$

[2 marks]

This indicates that the average amount of money they might expect to take in a day would be £414.70. [1 mark]

The standard deviation is £37.22, so around $\frac{2}{3}$ of the time they would expect to take between $414 - 37 = 377$ and $414 + 37 = 451$. [2 marks]

[9 marks]

- (b) The vendor is thinking about introducing a third flavour, and while they don't know what the mean sales for it will be, they have data of a sample of two weeks of daily sales from another vendor, which has a mean of 76 and an SD of 12.1. Give a 95% confidence interval for what the mean overall sales would be for this new flavour. (6 marks)

Solution: A 95% confidence interval for the population mean would be given by:

$$\left(\bar{x} - \left(1.96 \times \frac{\sigma}{\sqrt{n}} \right), \bar{x} + \left(1.96 \times \frac{\sigma}{\sqrt{n}} \right) \right)$$

Here $\bar{x} = 76$ and $\sigma = 12.1$, and $n = 14$. Then the confidence interval is

$$\left(76 - \left(1.96 \times \frac{12.1}{\sqrt{14}} \right), 76 + \left(1.96 \times \frac{12.1}{\sqrt{14}} \right) \right) = (76 - 6.33, 76 + 6.33) = (69.67, 82.33)$$

[6 marks]

Total marks for question 4: [23]